Orange juice or fructose intake does not induce oxidative and inflammatory response.

OBJECTIVE: We have previously shown that 300 kcal from glucose intake induces a significant increase in reactive oxygen species (ROS) generation and nuclear factor-kappaB (NF-kappaB) binding in the circulating mononuclear cells in healthy normal subjects. We hypothesized that the intake of 300 calories as orange juice or fructose, the other major carbohydrate in orange juice, would induce a significantly smaller response than that of glucose. RESEARCH DESIGN AND METHODS: Four groups (eight subjects each) of normal-weight subjects were given a 300-cal drink of glucose (75 g), fructose (75 g), or orange juice or water sweetened with saccharin (control group) to drink, and then blood samples were collected. RESULTS: There was a significant increase in ROS generation by mononuclear cells (by 130 +/- 18%, $P < 0.001$), polymorph nuclear cells (by 95 +/- 22%, $P < 0.01$), and in NF-kappaB binding in mononuclear cells by 82 +/- 16% ($P < 0.01$) over the baseline after 2 h of glucose intake. These changes were absent following fructose, orange juice, or water intake. There was significantly lower ROS generation and NF-kappaB binding following orange juice, fructose, and water compared with glucose ($P < 0.001$ for all). Furthermore, incubation of mononuclear cells in vitro with 50 mmol/l of the flavonoids hesperetin or naringenin reduced ROS generation by 52 +/- 7% and 77 +/- 8% ($P < 0.01$), respectively, while fructose or ascorbic acid did not cause any change. CONCLUSIONS: Caloric intake in the form of orange juice or fructose does not induce either oxidative or inflammatory stress, possibly due to its flavonoids content and might, therefore, represent a potentially safe energy source.